RURAL AI - UNIVERSITY OF NEWCASTLE AND UNIVERSITAS HASANUDDIN

SawahPintar

A soil sensor teaching tool for rice farming field workshops in South Sulawesi. One laptop, one probe, one focus group at a time.

Project team briefing, 30 July 2026. Use the arrow keys to move through this deck.

WHAT IT IS AND WHO IT IS FOR

A laptop that reacts to a probe in a farmer's hand

- Built for field workshops teaching Indonesian rice farmers scientific methods for raising crop yield.
- One laptop per focus group, driven by a Newcastle or Hasanuddin facilitator.
- Farmers get a hands-on moment: pushing an RS485 soil probe into the ground and watching the screen respond.
- Fully offline. No installation, no administrator rights, no internet connection during the workshop.
- Replication is copying a folder. Each set is independent and never talks to another laptop.

> The measure of success is whether a farmer leaves the session understanding one thing they could do differently in their own field.

THE PROBLEM

Farmers cannot see soil condition

Salinity, acidity and water status are invisible to the eye. A farmer judges a paddy by habit and by the crop's appearance days or weeks after a problem has already set in, not by a number checked in the moment.

Irrigation timing, salinity risk and fertiliser decisions are made without any direct measurement to weigh against experience. By the time a symptom shows in the plant, the window to act cheaply has often closed.

A tool that turns a soil property into something a standing group can watch change, in the seconds after a probe goes into the ground, gives that group a shared reference point the discussion can build on.

THE KIT

One laptop, one probe, one dongle

- An SN-3002-TR five-pin soil multi-parameter probe, RS485 Modbus RTU.
- One RS485-to-USB dongle per laptop, driver installed ahead of the workshop.
- A Windows laptop with Microsoft Edge, already present on every supported machine.
- Sets run independently. Four focus groups need four complete kits, not one shared server.

4800
baud, 8N1

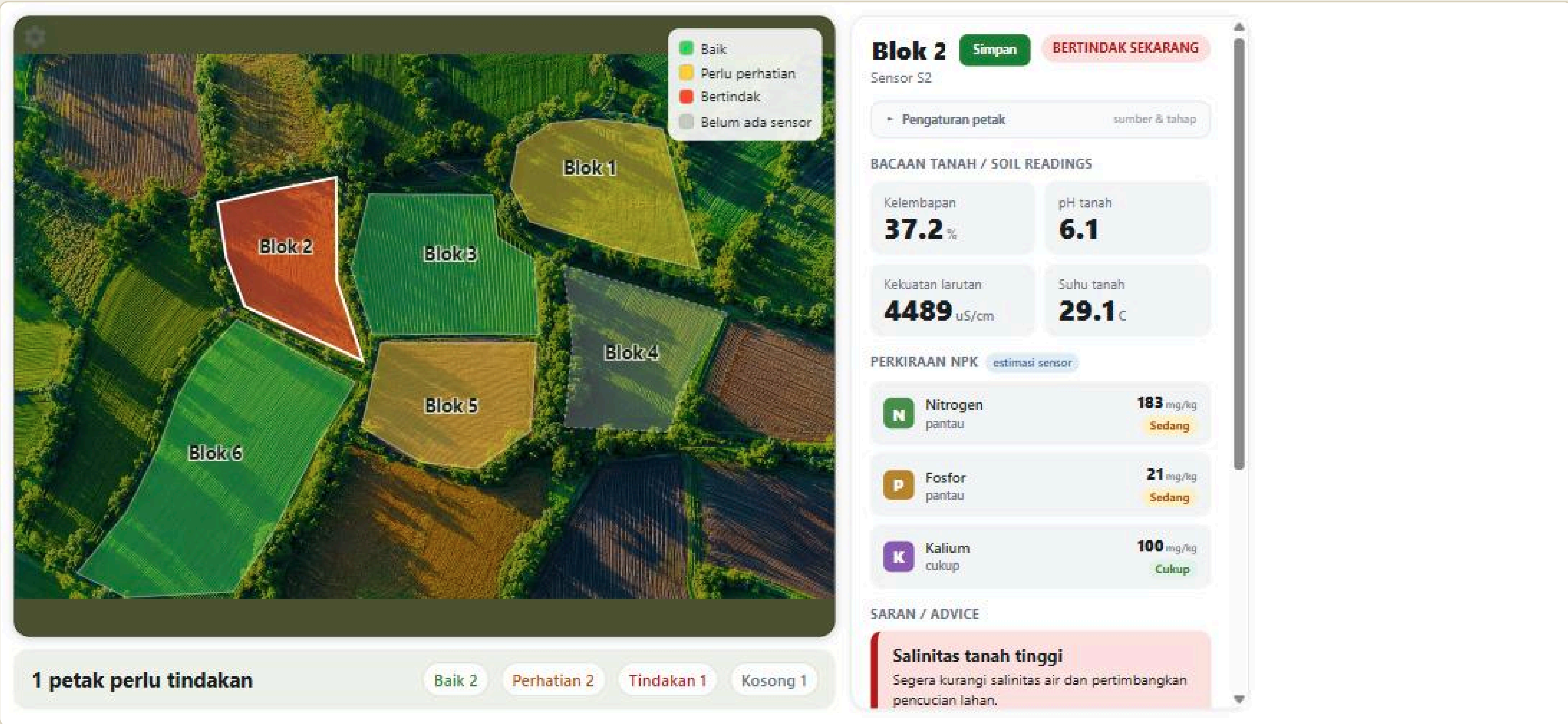
1
frame reads all 9 registers

0x01
slave address

Wiring: brown to power (4.5 to 30 V DC), black to ground, yellow to 485-A, blue to 485-B. Reversing A and B is the most common field mistake.

THE FARMER SCREEN

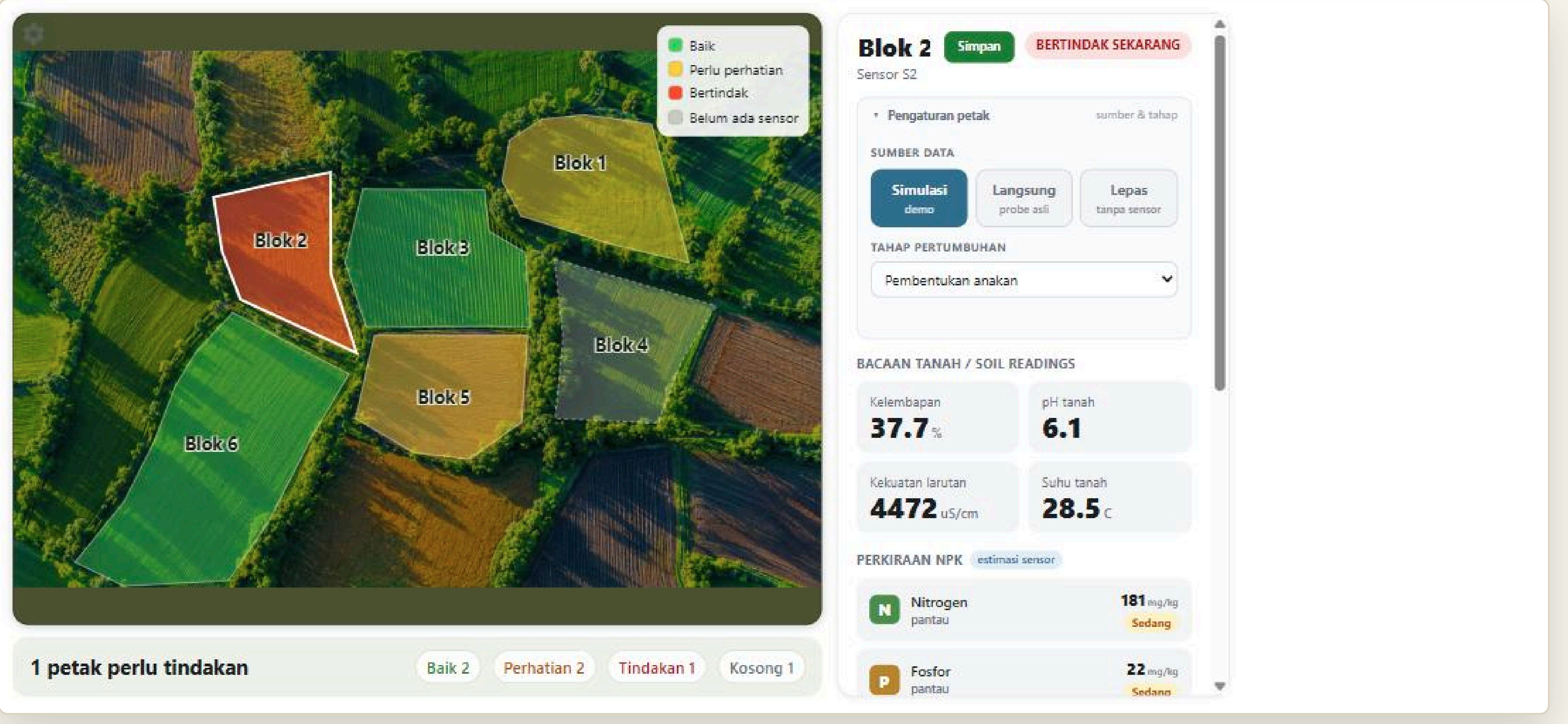
The field, drawn as a map of plots



One outlined plot per sensor, laid over the field and tinted by its alert: green Baik (satisfactory), amber Perlu perhatian (attention), red Bertindak (act now), and dashed grey Belum ada sensor for a plot with no sensor attached. A colour legend sits top-right; a summary bar below counts the plots (Baik / Perhatian / Tindakan / Kosong). It fits one screen with no scrolling, and there is no longer a SIMULASI watermark. Tapping a plot opens its detail on the right.

TAPPING A PLOT

The plot detail



An editable plot name with a Simpan button, and a collapsible Pengaturan petak panel: data source (Simulasi / Langsung / Lepas), a Port box when it is live, and a per-plot Tahap pertumbuhan growth-stage dropdown. Below sit the four soil readings (Kelembapan, pH tanah, Kekuatan larutan, Suhu tanah), a Perkiraan NPK estimate, and the advice cards.

THE FOUR ADVICE GROUPS

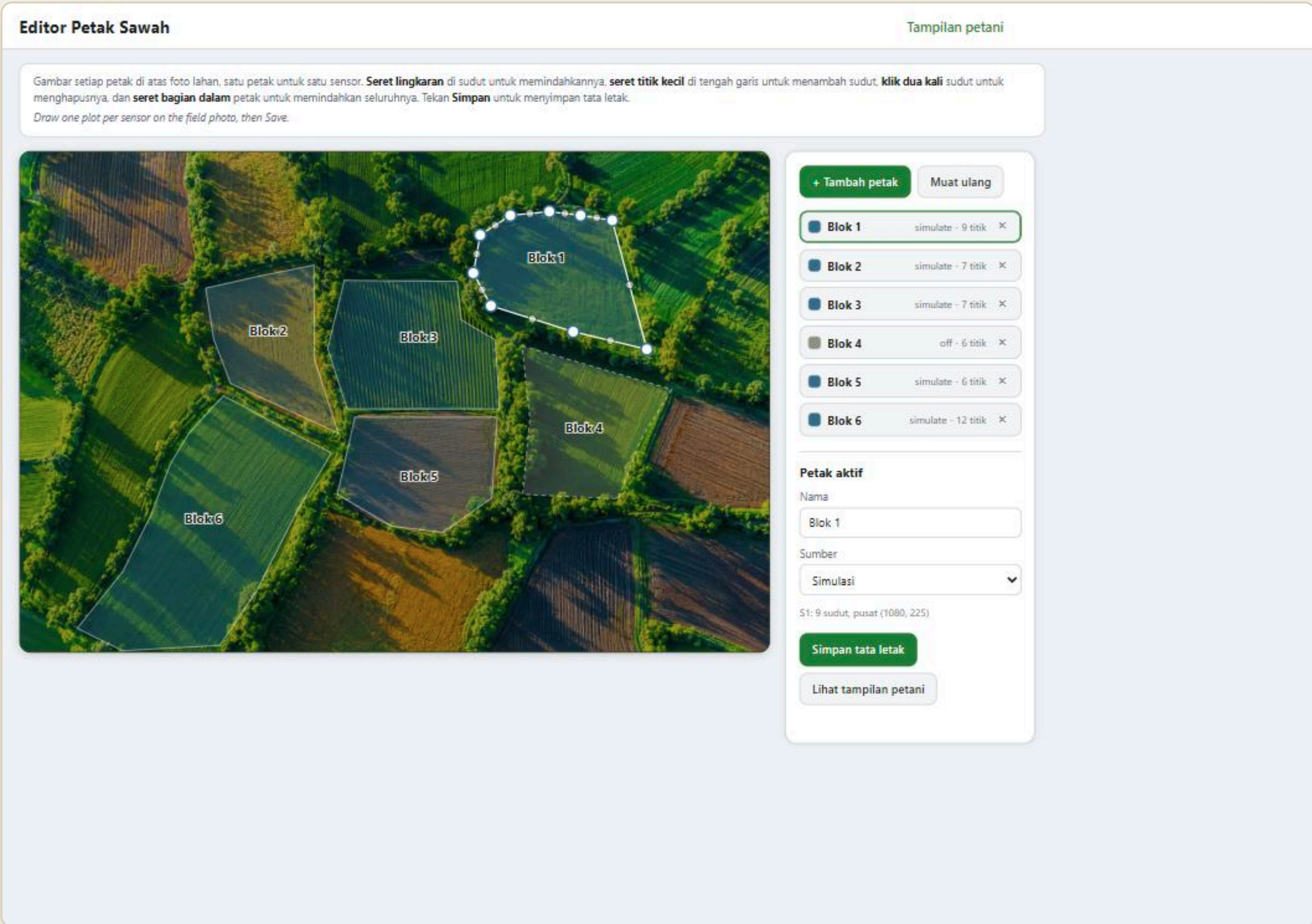
Every card traces back to a cited threshold

GROUP	RULE	PRIORITY
Salinity	Escalates one severity level during panicle initiation and flowering	Highest
Water (AWD)	Re-flood at about 15 cm water-table drawdown; suspended around flowering	Second
Acidity	Below pH 5.5, recommend lime or dolomite	Third
Nutrients	An NPK estimate ("estimasi sensor") in coarse bands, shown alongside the conductivity-derived soil-solution-strength indicator; a fertiliser dose still comes only from a PUTS test, never the probe	Fourth

Salinity sits above the other three because a saline event does the most damage the fastest, and rice is at its most sensitive to salt in the same window that acidity and nutrient advice change least. The nutrient group now surfaces an NPK estimate in coarse bands, clearly labelled as a sensor estimate rather than a dose; slide 11 explains the honesty position behind that choice.

SETTING UP THE FIELD

The zone editor



Open it from the top-left gear (or /zone-editor.html) and draw one plot per sensor over the field photo: drag a corner to move it, drag a mid-edge dot to add a corner, double-click a corner to remove it, drag the interior to move the whole plot. Set each plot's name, source and port, then Save persists the layout so the next launch opens straight on the map. The bundled photo is a stand-in; a real site swaps in a drone or satellite image and re-traces.

THE HANDS-ON MOMENT

The probe moment

The moment the whole workshop is built around still exists. One plot is set to **Langsung** (live) on its COM port, the farmer pushes the probe into the soil, and that plot reacts live on the map.

Its colour shifts and its readings update as the probe seats in wet soil - **Bertindak** to **Perhatian** in the seconds after a farmer's own hands go to work. A standing group watches one square on the field change, in real time, from something they did.

> A number on a screen is abstract. The same number, changing because you just pushed a probe into your own paddy, is not.

BEHIND THE MAP

Two setup surfaces, neither shown to farmers

The plot editor

The top-left gear now opens the **plot editor**, the main setup surface. This is where the field is drawn, one plot per sensor, and where each plot's name, source and port are set (slide 8).

The operator console

Reached from the plot editor's header link, or with **Ctrl+Alt+O**. It still holds the register inspector for unfamiliar sensors, the site name, the language toggle, the field card, PUTS class entry (the only path to a fertiliser dose), scenario rehearsal buttons, demo reset and data export.

THE HONESTY POSITION

Nitrogen, phosphorus and potassium are not three measurements

They are three functions of one conductivity reading. Three independent lines of evidence agree, and any of the three would be enough on its own:

1. The manufacturer's own datasheet footnotes describe those three registers as conductivity-derived values, not independent nutrient measurements.
2. The writable calibration registers apply only a scale and an offset to each output, giving three straight lines against one shared variable. No combination of coefficients can pull three nutrients back out of one input.
3. In air, the probe reads conductivity 0 with nitrogen, phosphorus and potassium all at 0 simultaneously. They move together because they are one number.

So the NPK values on each plot's detail are shown as an "estimasi sensor" - a deliberate choice for the demo, presented plainly as an estimate, never as a measurement or a dose. The conductivity-derived soil-solution-strength indicator remains alongside them, and a fertiliser dose still comes only from operator-entered PUTS status classes, via Permentan No. 13 of 2022.

WHAT THE EVIDENCE BASE IS

Every threshold in the rule engine traces to a source

CLAIM	SOURCE
Salinity is the top alert, stage-weighted, worst at panicle initiation and flowering	58-study meta-analysis, Zheng et al., European Journal of Agronomy, 2023
pH below 5.5 triggers lime or dolomite and a better P and K response	528 on-farm lowland rice trials in Indonesia, Field Crops Research, 2025
Safe AWD re-floods at 15 cm drawdown, suspended around flowering	IRRI Safe Alternate Wetting and Drying method
Fertiliser dose by nutrient status class	Permentan No. 13 of 2022, official Indonesian dose table

The salinity meta-analysis also reports a pooled mean yield loss of 64.5 per cent under salinity stress across its 58 studies. That figure is an experimental mean across often severe treatments, not a typical field outcome, and it is never shown to farmers for that reason. It belongs here, in front of this team, described exactly as what it is.

WHAT IS PROVISIONAL AND NEEDS FACULTY INPUT

Two kinds of draft, both marked in the files themselves

Content, not calibration

Every farmer-visible string in `data/content/` ships marked **draft: true**. Headlines, body text and icon choices are all placeholders awaiting Hasanuddin review, and may need Bugis or Makassarese terms rather than standard Bahasa Indonesia in places.

Calibration, not content

The mapping from bulk conductivity to a salinity class, and from volumetric moisture to an estimated water-table depth, in `data/calibration/default.yaml`, is marked **provisional: true**. Neither has been checked against local South Sulawesi paddy soils.

A length of perforated PVC pipe standing beside the probe during workshops teaches the genuine IRRI method and gathers the paired readings this calibration needs.

HOW A KIT IS BUILT

Verified end to end, not assumed

- Embeddable Python 3.12.8 from python.org, 11,094,114 bytes, runs after extraction with no installer and no administrator rights.
- **The trap:** the shipped `python312._pth` has `#import site` commented out. Left that way, every dependency import fails even though pip reports success. One uncommented line is the difference between a working kit and a dead one.
- **pytz is required and non-obvious:** duckdb needs it to read TIMESTAMPTZ columns but does not declare it. Omitting it fails only on a clean machine, which is every workshop laptop. Running the suite against the real runtime broke 11 tests before this was caught.
- **websockets is required and easy to miss:** uvicorn needs it to upgrade the live /ws feed that pushes readings to the map. Without it the socket never opens and the live map never updates, even though the app starts cleanly.
- Seven pure-wheel dependencies in total: duckdb, pyserial, fastapi, uvicorn, websockets, pyyaml, pytz.
- Assembled runtime with dependencies: about 92 MB on disk.
- **Certification rule:** always build the runtime from scratch and run the full test suite against it. Never certify a kit from a developer machine.

WHAT WE NEED FROM THE TEAM NEXT

Three concrete inputs, all faculty-side

- **Bahasa Indonesia wording review.** Every headline, body line and icon choice in `data/content/` is a draft. Hasanuddin faculty sign-off turns these into farmer-ready text, including any Bugis or Makassarese terms that fit the workshop sites better than standard Bahasa Indonesia.
- **Whether PUTS kits are available at each site.** The fertiliser layer is materially stronger with a PUTS reading on hand; the design works either way, but we need to know which sites have a kit.
- **Local calibration values.** Paired readings of bulk conductivity against a saturated paste extract, and moisture against a perforated field tube, for the soils where the workshops will actually run.